Definitions and responsibilities

DPS	Detector Protection System: arbitrates protection, applies permits and inhibits, and verifies the equipment response.
SC	Slow controls, using Ignition: operates and monitors equipment, coordinates requests, and owns the logs.
DAQ	Configures readout, takes data, and uses detector availability and activity from SC.
BMS / cryo	Building Management Systems and cryogenic controls own their services and provide agreed states to SC.
Subsystems	Define operating limits, required protection and allowed response times.
Local interlocks	Equipment-level protection: can trip without SC or the central DPS.

An operating request cannot override a protection trip.

Personnel-safety alarms remain with SDSTA. Agree equipment-protection responsibility for each phase when DPS is not active.

Facility information reaches DPS through SC

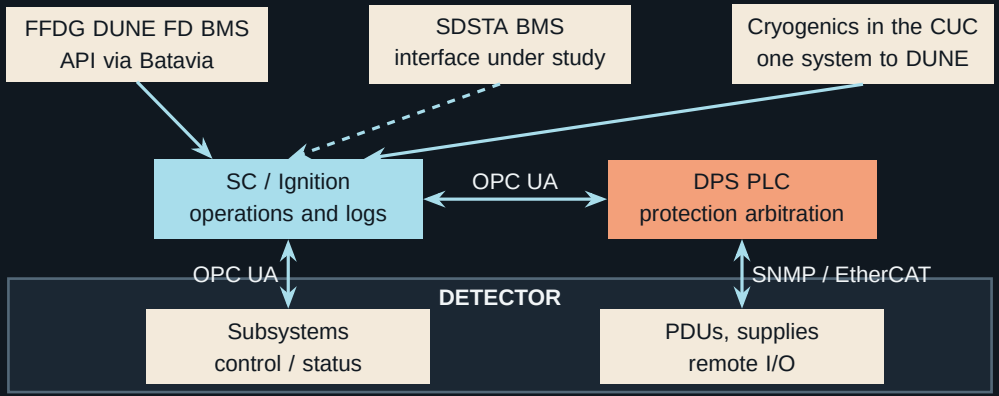
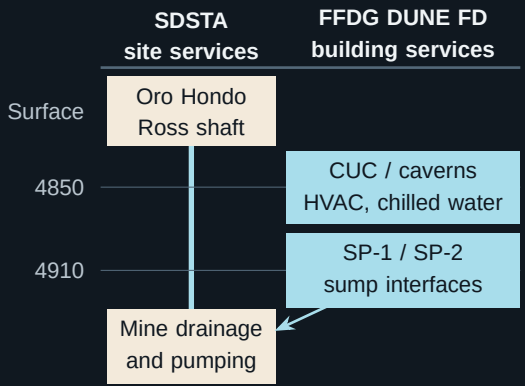


Diagram. SC connects facility and detector information; DPS decides and applies the protection action.

EtherCAT connects remote I/O; wired inhibits connect that I/O to the equipment. SDSTA exchange is under investigation. Proposed central DPS PLC location: CUC.

FFDG DUNE FD BMS: the Metasys API route



ISD proposal

Metasys API in Batavia interfaces the FFDG BMS to DUNE FD SC.

SBN exercise

The API has been exercised with SBN building data and an OPC UA server for experiments in that building.

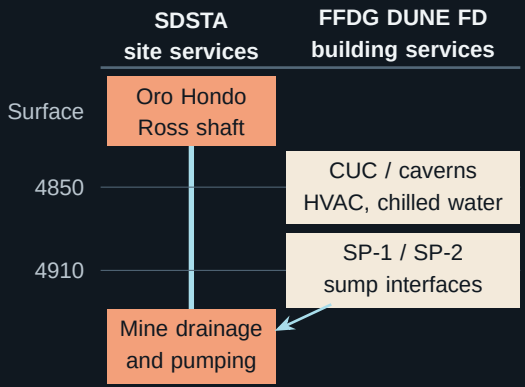
Still missing

The licence for live data. Historical data and OPC UA publication have been exercised.

Diagram. DUNE building services at 4850/4910 connect to wider SURF services. Schematic, not to scale.

U-FD-M-901 rev12 and BSI section 230923; API status: Manuel and October test records. Cryogenic process controls are separate from BMS.

SDSTA BMS: survey the hardware and agree the exchange



Hardware survey ongoing
Identify controllers, I/O, available contacts and network/fibre ports.

Protocols under investigation
Identify exported states and the path to SC.

Agreement options
Agree ownership, allowed exchanges, fault reporting and maintenance.

Diagram. SDSTA site services extend beyond the DUNE caverns; BMS connection points remain under survey.

4850/4910 maps are ready; access work involves Rachel and SDSTA integrator Dale Curran. Controller models and BMS points remain under survey.

Cryogenics: one system to SC and DPS

Cryogenics is in the CUC. Its controls are divided internally; DUNE receives one coherent system interface.

Topic	Status and detector consequence
Process control	Cryogenics owns regulation and its process/safety controllers; the architecture is documented.
Critical conditions	Low LAr level, relief-valve opening and defined boiling-risk conditions affect HV or CRP-bias permits.
Interface	Cryogenics $\rightarrow$ SC $\leftrightarrow$ DPS. Name the source of each measurement; some level signals belong to detector instrumentation.
Running agreement	Agree readiness states, limits, signal quality, loss response and authorization for each phase.

Cryogenic architecture and 2022 FDR describe the control system. They do not establish an agreed detector run-permit contract. HV/CRP actions come from the interface requirements under review.

Start with the interlocks proposed by each consortium

The matrix collects the current ICD requirements and consortium proposals. We review each system against those proposed interlocks.

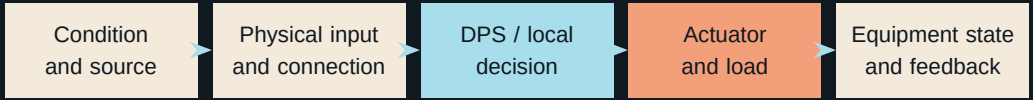


Diagram. Trace the physical and control path, including the time from detection to completed shutdown.

For each system: identify the hardware already provided, missing connections or components, and requirements that change the hardware choice.

Use the existing rack lists, drawings, parts and power calculations. Count shared I/O once. Proposed actions and unresolved conflicts remain visible in the matrix.

The DPS brain turns conditions into equipment permits

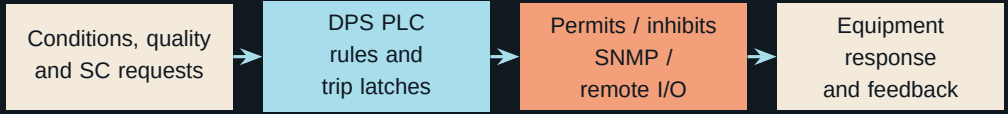


Diagram. SC requests an action; the PLC applies the protection rules and reports the result.

Enable: request present, required conditions valid, no blocking trip.

Trip: remove the affected permit, retain the cause and verify the equipment state. SC records the event.

Recovery: reset and re-enable are separate actions.

Proposed logic on a commercial PLC based on BDE. Define stale-input and communication-loss responses per load. Local equipment interlocks remain active.

Detector

BDE: the existing VD design sets the hardware baseline

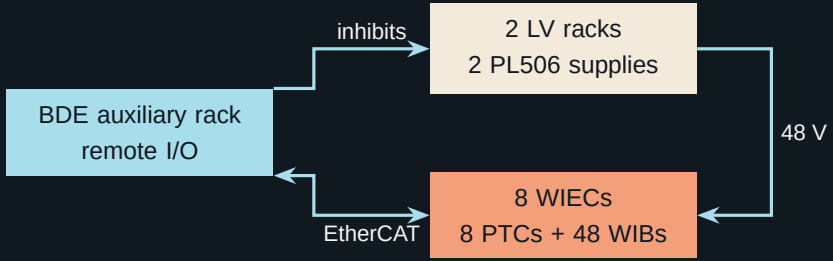


Diagram. Ten groups: 10 auxiliary racks, 20 LV racks, 20 PL506 supplies, 80 WIECs and 480 WIBs.

The panel drawings, parts lists, power calculations and Ignition screens are available. Ten bias racks provide the separate CRP/FC protection interfaces.

VD rack workbook and June SC layout. BDE = bottom drift electronics; WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. Design/test evidence is not a claim of VD installation.

BDE trip example: fan cooling and the bias dependency

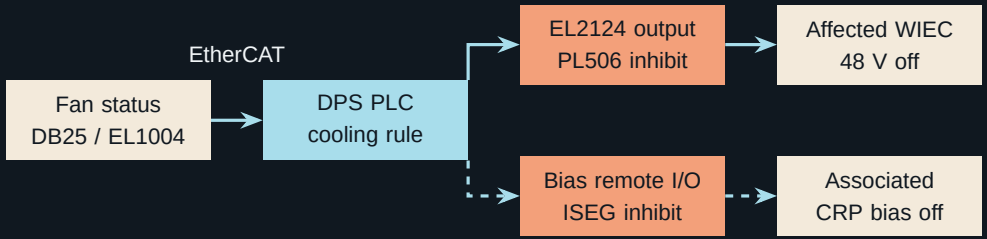


Diagram. Solid: documented BDE cooling path. Dashed: associated bias inhibition, if required by the VD rule.

The output paths already exist. Map the affected WIEC and bias channels, set the cooling rule, and verify supply voltage/state after the trip.

The historical HD rule links cooling and bias. The VD cooling proposal currently assigns LV inhibition; its bias dependency needs agreement. Six fan-status inputs per WIEC imply 48 per eight-WIEC group, not 48 new inputs.

PDS DAPHNE: control, power and the DAQ interface

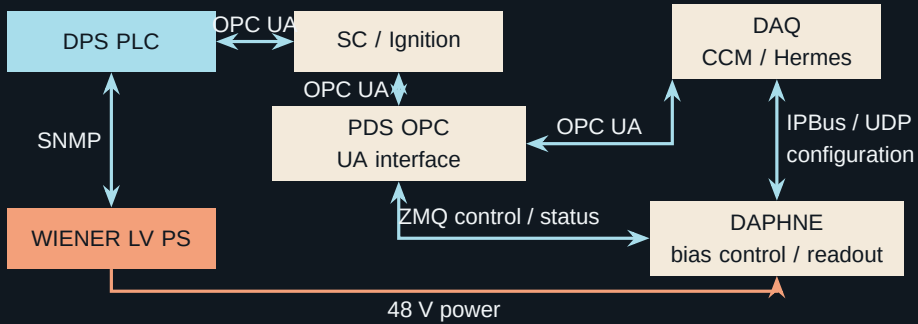


Diagram. Working ICD DAPHNE paths plus DPS and LV control. PDS power scope: 34 LV feeds.

SiPM-bias inhibit: DPS → SC → DAPHNE. Bias state returns to SC; DAQ receives detector availability.

DAPHNE-controlled bias only; cathode power follows PoF. Test inhibit acknowledgement and timing.

PDS PoF: monitor locally, inhibit through the PDU

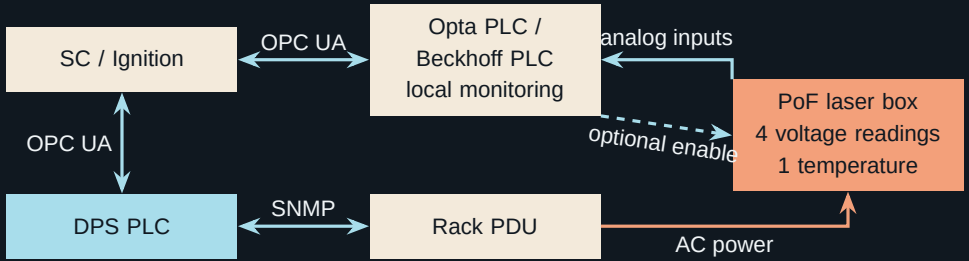


Diagram. Selected inhibition route: switch the mapped PDU outlet. Explore a separate box-enable input if available.

80 boxes: 320 voltage + 80 temperature readings = 400 readings. Size each controller option for these physical inputs and its box grouping.

Four voltage readings and one chassis temperature per box: Manuel's current scope. Readings do not count laser drivers or central DPS terminals. The optional control signal and feedback remain to be identified.

HV: drift voltage and CRP / field-cage bias

VD inventory: one Heinzinger drift-HV supply; ten ISEG bias crates, 576 channels = 480 CRP + 96 field-cage channels.

Equipment	Control / monitoring	Protection path
Drift HV	SC / OPC UA ↔ LabVIEW / cRIO; analog setpoint and V/I readback	DPS → EtherCAT remote I/O → wired Heinzinger interlock
CRP / FC bias	SNMP supply control/status; SC receives states and logs through DPS	DPS → EtherCAT remote I/O → wired ISEG channel inhibits

Example: low LAr level removes the drift-HV permit. A drift-HV trip also shuts down the associated field-cage termination supplies.

EDMS 3315615 v1 attachment, internal v2; October architecture. Map actual pins, channels and feedback before counting added I/O. Supply shutoff and voltage decay are separate timing requirements.

CalCI temperature sensors and their readout

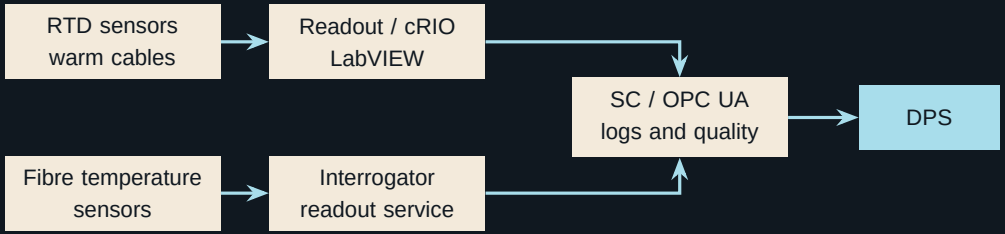


Diagram. Detector instrumentation is separate from cryogenic process control. Name the sensor and its physical location.

Temperature profiles support cooldown and operation. A protection rule needs the affected equipment, threshold, persistence and allowed response time.

Hardware review: match RTD / fibre channels, warm readout and service interfaces to the required measurements.

October SC architecture and CalCI inventory. Trace level measurements to their own instrument and readout.

CalCI lasers: coordinate emission and detector operation

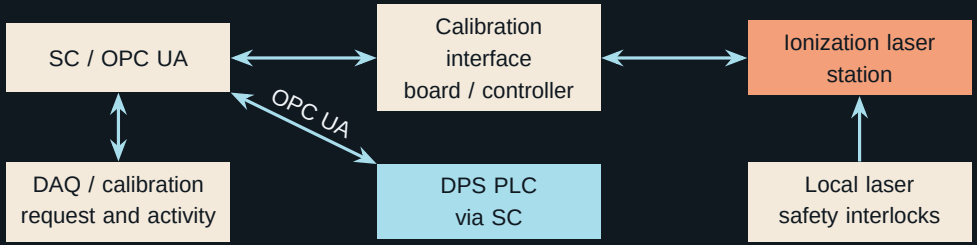


Diagram. Working inventory: four ionization-laser stations. The CIB has an OPC UA interface in the SC architecture.

Proposed operating rule: deny conflicting laser/PDS requests. Require emission or shutter-off confirmation before PDS return.

Hardware review: station power, permits, shutter/emission feedback and trigger connection.

CalCI inventory; October SC architecture; proposed rule O-002. Local personnel-safety interlocks stay independent.

CalCI purity monitors: separate activity from protection

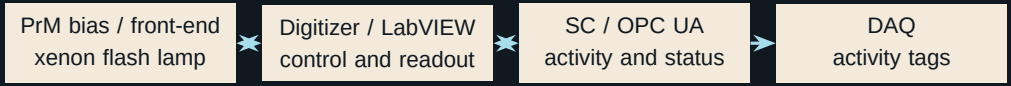


Diagram. Track lamp activity, PrM supply state and the affected detector region.

Proposal Q-001: allow acquisition and publish the activity interval for DAQ tagging.

Open ICD conflict: the older PDS ICD requests SiPM bias off while the purity monitor is active. The consortium must agree which rule applies.

Hardware review: lamp control/status, HV/LV supplies, digitizer interfaces and the readback used to verify the action.

Working PrM inventory; EDMS 3309688 v1 section 5; SC–PDS–DPS working matrix. No automatic bias trip is assigned by Q-001.

Racks: one protection package shared by the subsystems

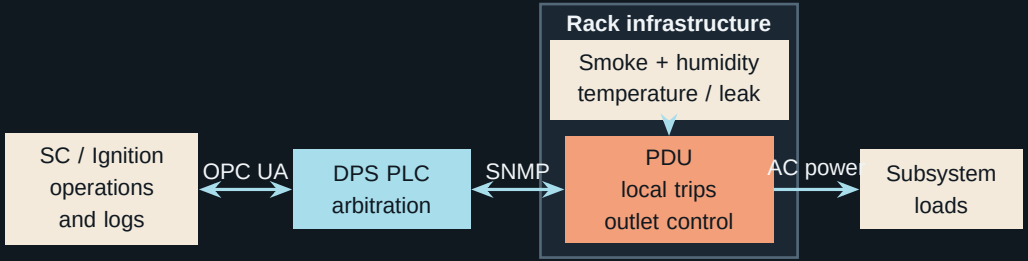


Diagram. Count the rack, PDUs, sensors and interface parts once. Link each subsystem to its PDU outlets.

Local PDU rules protect the mapped outlets. Humidity is monitored; a humidity shutdown needs a stated requirement.

Rack inventory includes 20 TDE racks: current DPS control is through their PDUs. Crate status and external inhibit remain future options. Reset and re-enable are separate actions.

Test arbitration, trips and recovery on the bench

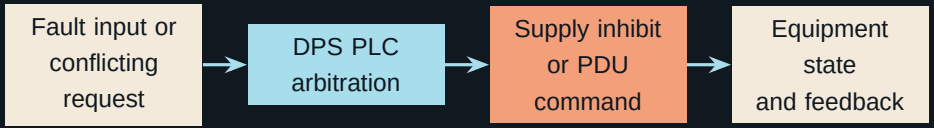


Diagram. Use the BDE vertical-slice work and G271 equipment for representative VD paths.

Measure detection through completed shutdown. Test local trips, SC and fieldbus loss, reset and re-enable. Verify that SC logs the decisions and equipment states.

No critical sub-ms function has been identified so far. Simulation prepares the fault and load cases; hardware tests measure the complete response.

Take the VD hardware to review in June 2027

Work before review	Result for procurement
Trace the VD inhibits	Existing hardware, added channels and cables, quantities and open alternatives
Build representative paths	Measured trip time, failure/recovery results and SC logging
Prepare purchases	Panel space and power, drawings, parts lists, costs, lead times and spares

After readiness review: order the hardware and install it in the site sequence.

Possible facility-interface need: summer 2027. Full VD response-matrix review: early 2028. The current path review establishes the requirements that affect hardware purchases.

I will lead the VD hardware work with a control engineer

Proposed contribution	Work
Manuel + control engineer	VD design, SURF interfaces and procurement package
Terri and Trevor	BDE panels, PLC/I/O implementation and bench work
Geoff + Manuel	SC commands, PDU protection and recovery
Vivian + consortia / DPS	Equipment requirements and ICD updates

Support needed: control-engineering time and access to the existing test stands.

SURF work: inspection maps are ready; access is being coordinated with Rachel, with Dale Curran identified as an SDSTA integration contact.

BDE drawings, parts and software are already available

TPC protection panel designs

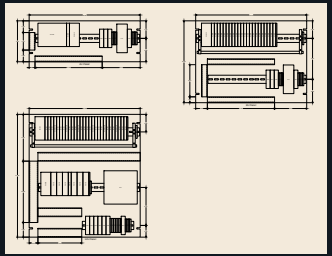
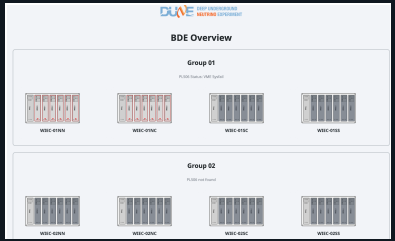


Diagram. Central PLC, bias I/O and LV/fan/heater reference panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

Historical TPC reference: EDMS 2401090 v4, sheet 2, with parts and power workbooks. The VD baseline is eight WIECs per group.

BDE cooling already has an input-to-inhibit path

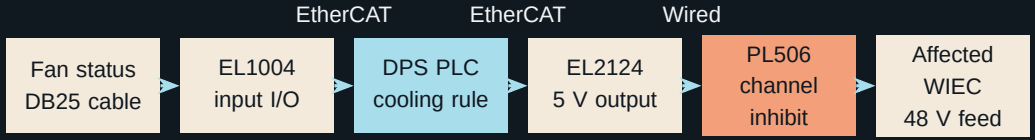


Diagram. Cooling fault → withdraw the permit for the affected PL506 channel.

Feedback: PL506 state and voltage/current → SNMP → DPS; OPC UA carries the result to SC for logging.

VD allocation check: six fan-status signals per WIEC means 48 per eight-WIEC group. Match these to the populated VD inputs and cables.

The six-fan WIEC note differs from the four-fan ICEBERG reference. These are baseline requirements, not new-module counts. Fan voting and allowed shutdown time remain to be agreed.

Backup: DAPHNE paths in the working ICD

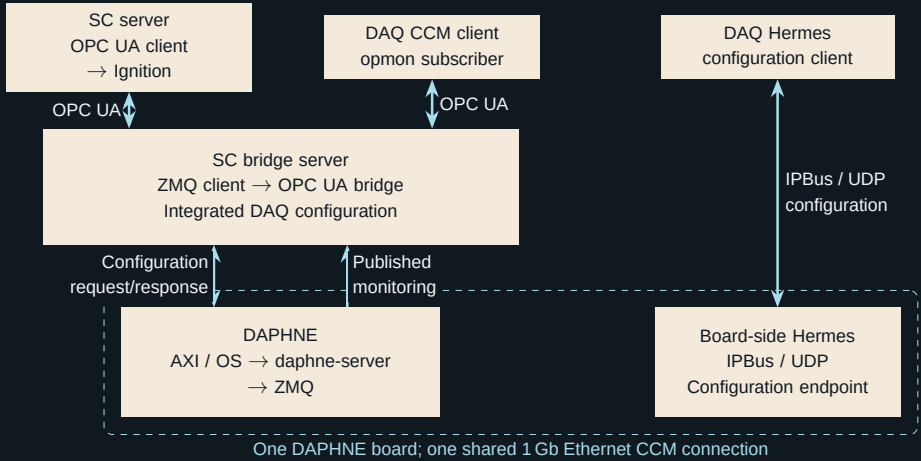


Diagram. Working SC–PDS–DPS ICD, Figure 1. DAQ CCM/opmon and Hermes remain distinct clients.

Backup: VD rack space and the existing DPS allocation

Rack type	Count	DPS allocation and workbook space
BDE auxiliary	10	12U DPS; U3–U21 marked Space (19U); eight WIECs per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U DPS; U3–U12 marked Space (10U)

Space marked in the rack workbook is not a new DPS reservation. Free DIN-rail length and I/O capacity depend on the populated VD panel.

May 2025 VD workbook: BDE Aux Rack and Bias HV Rack. Auxiliary U9 also has a possible ground-plane readout note. U is rack height.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. This is the first-pass layout; the difference from 576 bias channels is not verified spare capacity.

Use the component loads for the VD power budget

Equipment	Power / capacity	Basis and limit
PL506 input	2.25 kW each	20 supplies: about 45.1 kW expected AC input
WIEC fans only	380 W per group	$8 \times 6 \times 0.33\,\text{A} \times 24\,\text{V}$; excludes heaters and I/O
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The inherited 831 W auxiliary figure needs a VD component budget covering fans, heaters and populated I/O.

May workbook; WIEC safety note, internal revision 0. Fan arithmetic assumes six fans per WIEC. Rated load, expected AC input and DC supply capacity are distinct.

Rack infrastructure already has a hardware reference

Rack component	Reference hardware / allocation
PDU	Raritan PX3; inlet and outlet allocation for the connected loads
Smoke detector	4WTAR-B and EOLR-1 end-of-line relay
Contact interface	DX-PD2C5; 12 V supply to the smoke detector
Humidity	Record the probe and peripheral interface for each rack
Temperature / water leak	PDU peripheral sensors where fitted

The May workbook includes a 76-entry rack-protection/PDU allowance. Map the actual rack packages against this existing allowance.

DUNE-doc-32655 v8, sections 3 and 5; May workbook, Standard Rack and Power Summary!B28. The guide does not specify a humidity probe model or trip requirement.

Reference: local PDU trips and recovery

PX3 setup guide	Local behaviour documented
Smoke, water, temperature	Local event rules and scripts can switch off the protected outlets
Network-switch outlet	Kept powered during the guide's fallback shutdown
After a local trip	Power is restored deliberately; no automatic restoration

The VD control chain is Ignition → DPS PLC → PDU. Local trip status returns to the PLC and to SC for logging.

DUNE-doc-32655 v8 describes an Ignition-direct prototype with local fallback on loss of Ignition. The VD PLC command path requires matching reachability, trip and reset configuration; it is not a claim that the prototype already implements it.

The VD inventory covers the other subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	34 LV feeds in the current presentation scope; supply status and control interfaces
Power over fibre (PoF)	80 boxes; 320 voltage and 80 temperature readings; Opta PLC or Beckhoff PLC alternative
TDE	20 mezzanine racks; PDU control now; other TDE interfaces are possible extensions
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

Subsystem exchanges use OPC UA through SC. PoF uses the current 4 V + 1 temperature case; these readings do not count central DPS terminals.

The BMS drawing identifies 20 control-panel symbols

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Inspection planned with Rachel; Dale Curran is the SDSTA integration contact identified for this work.

U-FD-M-901 rev12, 14 February 2025; marked 4850/4910 maps and survey register. Symbols are drawing references; field observations are pending.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 allocation: rack positions, distinct from PLC panels and remote-I/O stations. Includes the listed reserved and spare positions.